CHAPTER 3

# System design

AERA is a digital twin of the LRT Kelana Jaya Line. It runs every train on the published timetable at real Kuala Lumpur time and predicts how crowded each train, and each car, will be. Predictions come from twelve weeks of origin-destination ridership published on data.gov.my, shaped by hour of day.

## 3.1 Crowding model

Each train's load is the passenger flow on its track segment multiplied by the scheduled headway and divided by the train's capacity of 800 riders. The result is shown as one of five crowding levels, always with a label and an icon so it never relies on colour alone.

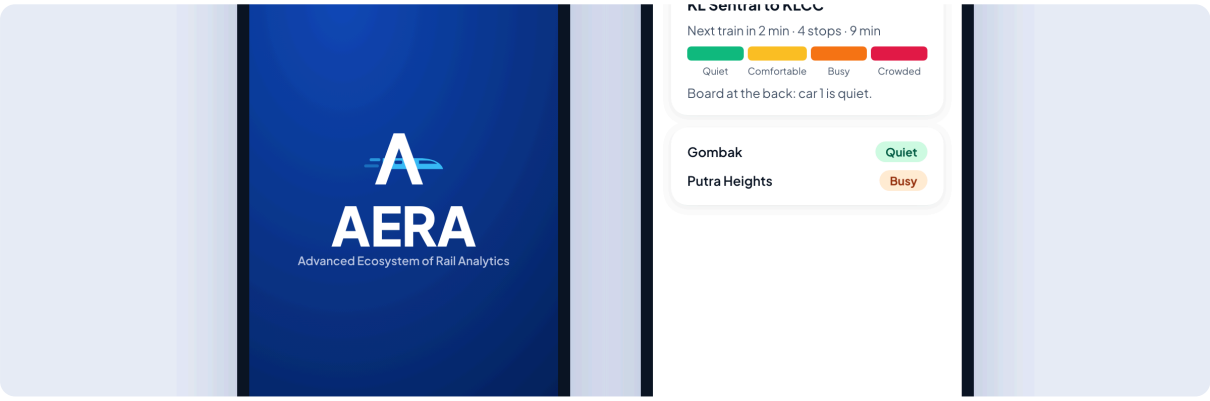



Figure 3.1: Splash screen and app bar in the AERA rider app.

| Level | Load | What riders see |
|---|---|---|
| Quiet | under 20% | Plenty of seats |
| Comfortable | 20 to 50% | Some seats, easy standing |
| Busy | 50 to 75% | Standing, room to move |
| Crowded | 75 to 95% | Tight standing |
| Packed | 95% and over | Consider the next train |

The live feed blends rider reports and in-train sensors into the model where they exist, weighted by how recent and how reliable each source is.

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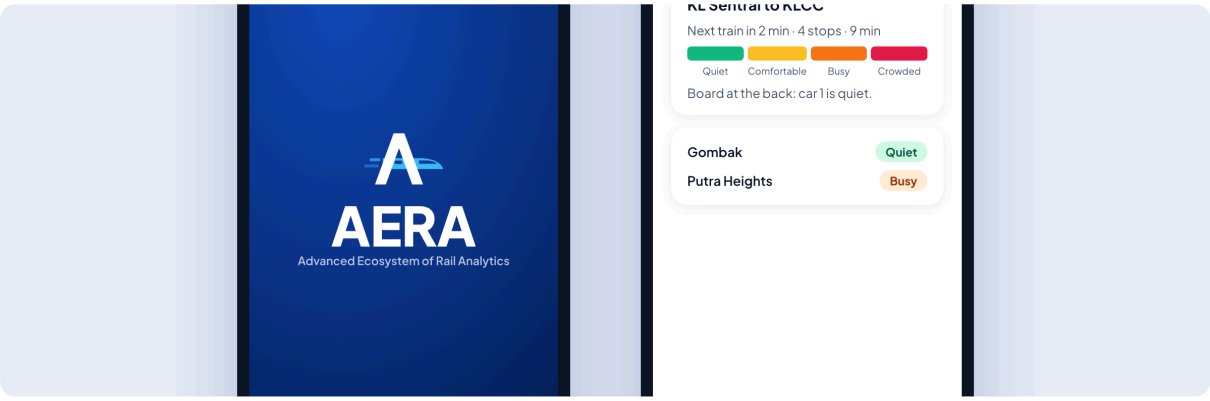



Figure 3.2: Splash screen and app bar in the AERA rider app.

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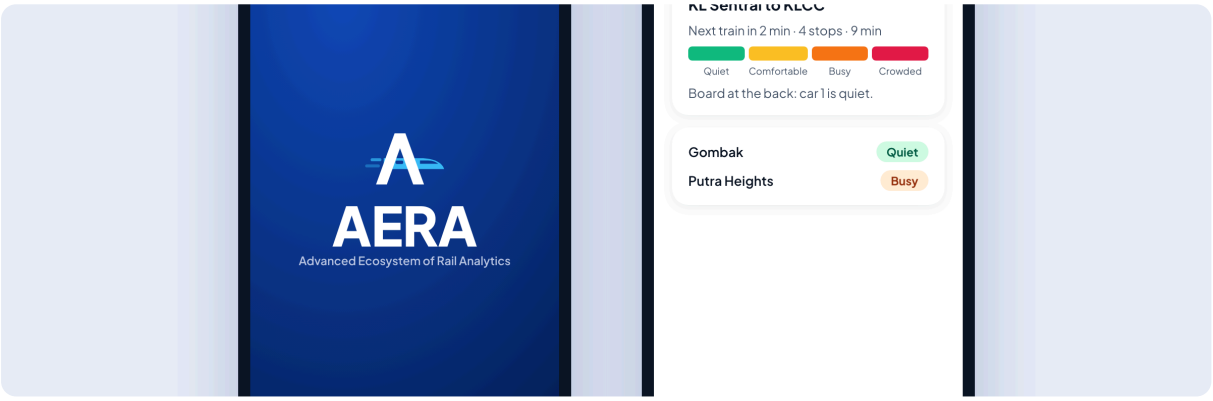



Figure 3.3: Splash screen and app bar in the AERA rider app.

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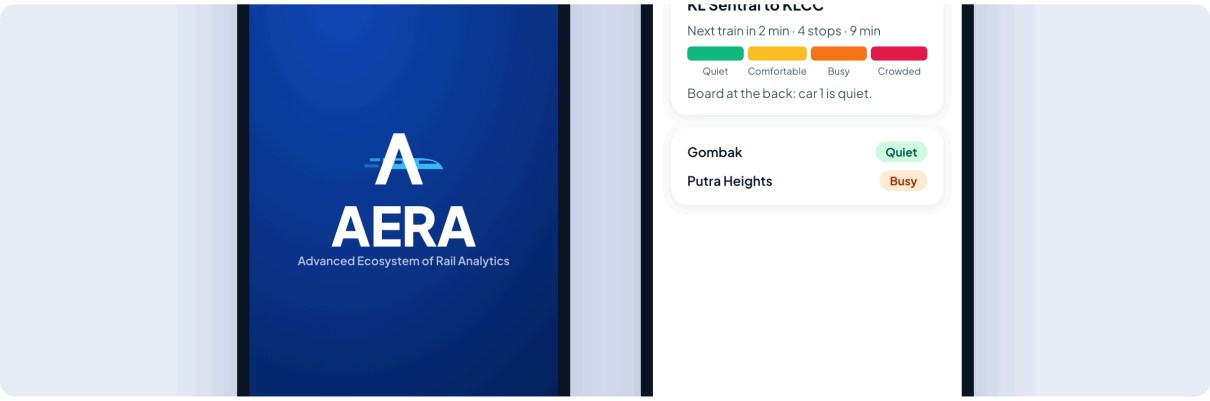



Figure 3.4: Splash screen and app bar in the AERA rider app.

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